



网络安全实验

防火墙



01

访问控制部分

- 设置iptables规则，实现指定功能
- 设置iptables规则，实现特定远端客户SSH连接

02

NAT部分

- 设置iptables规则，实现内网服务的发布



01

访问控制部分

- 设置iptables规则，实现指定功能
- 设置iptables规则，实现特定远端客户SSH连接



✓ 防火墙

Kali

✓ 其他主机

Ubuntu

✓ 软件

iptables、wireshark

查看主机信息

使用命令ifconfig查看主机地址信息。两虚拟机之间应能互相ping通。

```
(root@kali)-[~]
# ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.133.136 netmask 255.255.255.0 broadcast 192.168.133.255
    inet6 fe80::94e4:385a:c603:9d16 prefixlen 64 scopeid 0x20<link>
    ether 00:0c:29:96:4a:12 txqueuelen 1000 (Ethernet)
    RX packets 322 bytes 50277 (49.0 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 301 bytes 46240 (45.1 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

```
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 4 bytes 240 (240.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 4 bytes 240 (240.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

```
(root@kali)-[~]
# ping 192.168.133.134
PING 192.168.133.134 (192.168.133.134) 56(84) bytes of data.
64 bytes from 192.168.133.134: icmp_seq=1 ttl=64 time=1.46 ms
64 bytes from 192.168.133.134: icmp_seq=2 ttl=64 time=0.521 ms
64 bytes from 192.168.133.134: icmp_seq=3 ttl=64 time=0.500 ms
^C
--- 192.168.133.134 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2018ms
rtt min/avg/max/mdev = 0.500/0.826/1.458/0.446 ms
```

```
ens33: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.133.134 netmask 255.255.255.0 broadcast 192.168.133.255
    inet6 fe80::faed:fcf8:4ed:4207 prefixlen 64 scopeid 0x20<link>
    ether 00:0c:29:f0:06:e4 txqueuelen 1000 (Ethernet)
    RX packets 286530 bytes 375603684 (375.6 MB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 66044 bytes 5747615 (5.7 MB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

```
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 23847 bytes 2075951 (2.0 MB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 23847 bytes 2075951 (2.0 MB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

```
ubuntu@ubuntu-virtual-machine:~$ ping 192.168.133.136
PING 192.168.133.136 (192.168.133.136) 56(84) bytes of data.
64 bytes from 192.168.133.136: icmp_seq=1 ttl=64 time=3.53 ms
64 bytes from 192.168.133.136: icmp_seq=2 ttl=64 time=0.470 ms
64 bytes from 192.168.133.136: icmp_seq=3 ttl=64 time=0.435 ms
^C
--- 192.168.133.136 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2008ms
rtt min/avg/max/mdev = 0.435/1.478/3.529/1.450 ms
```



iptables

禁止所有主机ping本地主机

命令示例：iptables -A INPUT -p icmp -j DROP

```
(root@kali)-[~]  
# iptables -A INPUT -p icmp -j DROP
```

```
ubuntu@ubuntu-virtual-machine:~$ ping 192.168.133.136  
PING 192.168.133.136 (192.168.133.136) 56(84) bytes of data.  
64 bytes from 192.168.133.136: icmp_seq=1 ttl=64 time=3.53 ms  
64 bytes from 192.168.133.136: icmp_seq=2 ttl=64 time=0.470 ms  
64 bytes from 192.168.133.136: icmp_seq=3 ttl=64 time=0.435 ms  
^C  
--- 192.168.133.136 ping statistics ---  
3 packets transmitted, 3 received, 0% packet loss, time 2008ms  
rtt min/avg/max/mdev = 0.435/1.478/3.529/1.450 ms  
ubuntu@ubuntu-virtual-machine:~$ ping 192.168.133.136  
PING 192.168.133.136 (192.168.133.136) 56(84) bytes of data.  
^C  
--- 192.168.133.136 ping statistics ---  
8 packets transmitted, 0 received, 100% packet loss, time 7174ms
```

可以ping通



不能ping通

仅允许某特定IP主机ping本地主机

命令示例：iptables -I INPUT -p icmp -s 192.168.133.134 -j ACCEPT

```
(root@kali)-[~]  
# iptables -L  
Chain INPUT (policy ACCEPT)  
target prot opt source destination  
ACCEPT icmp -- 192.168.133.134 anywhere  
DROP icmp -- anywhere anywhere  
  
Chain FORWARD (policy ACCEPT)  
target prot opt source destination  
  
Chain OUTPUT (policy ACCEPT)  
target prot opt source destination
```

```
ubuntu@ubuntu-virtual-machine:~$ ping 192.168.133.136  
PING 192.168.133.136 (192.168.133.136) 56(84) bytes of data.  
^C  
--- 192.168.133.136 ping statistics ---  
8 packets transmitted, 0 received, 100% packet loss, time 7174ms  
  
ubuntu@ubuntu-virtual-machine:~$ ping 192.168.133.136  
PING 192.168.133.136 (192.168.133.136) 56(84) bytes of data.  
64 bytes from 192.168.133.136: icmp_seq=1 ttl=64 time=0.447 ms  
64 bytes from 192.168.133.136: icmp_seq=2 ttl=64 time=0.451 ms  
64 bytes from 192.168.133.136: icmp_seq=3 ttl=64 time=0.421 ms  
^C  
--- 192.168.133.136 ping statistics ---  
3 packets transmitted, 3 received, 0% packet loss, time 2044ms  
rtt min/avg/max/mdev = 0.421/0.439/0.451/0.013 ms
```

不能ping通



再次ping通



iptables

允许每10s通过一个ping包

命令示例 : `iptables -I INPUT -p icmp -m limit --limit 6/min --limit-burst 1 -j ACCEPT`

`iptables -A INPUT -p icmp -j DROP`

```
(root@kali)-[~]
# iptables -L
Chain INPUT (policy ACCEPT)
target     prot opt source                destination          limit: avg 6/min burst 1
ACCEPT     icmp -- anywhere             anywhere
DROP       icmp -- anywhere             anywhere

Chain FORWARD (policy ACCEPT)
target     prot opt source                destination

Chain OUTPUT (policy ACCEPT)
target     prot opt source                destination
```

```
ubuntu@ubuntu-virtual-machine:~$ ping 192.168.133.136
PING 192.168.133.136 (192.168.133.136) 56(84) bytes of data.
64 bytes from 192.168.133.136: icmp_seq=1 ttl=64 time=0.704 ms
64 bytes from 192.168.133.136: icmp_seq=11 ttl=64 time=0.603 ms
64 bytes from 192.168.133.136: icmp_seq=21 ttl=64 time=0.555 ms
^C
--- 192.168.133.136 ping statistics ---
21 packets transmitted, 3 received, 85.7143% packet loss, time 20480ms
rtt min/avg/max/mdev = 0.555/0.620/0.704/0.062 ms
```

阻断来自某MAC地址的数据包

命令示例 : `iptables -A INPUT -m mac --mac-source <your_MAC> -j DROP`

```
(root@kali)-[~]
# iptables -A INPUT -m mac --mac-source 00:0c:29:f0:06:e4 -j DROP

(root@kali)-[~]
# iptables -L
Chain INPUT (policy ACCEPT)
target     prot opt source                destination          MAC 00:0c:29:f0:06:e4
DROP       all  -- anywhere             anywhere

Chain FORWARD (policy ACCEPT)
target     prot opt source                destination

Chain OUTPUT (policy ACCEPT)
target     prot opt source                destination
```

```
ubuntu@ubuntu-virtual-machine:~$ ping 192.168.133.136
PING 192.168.133.136 (192.168.133.136) 56(84) bytes of data.
^C
--- 192.168.133.136 ping statistics ---
6 packets transmitted, 0 received, 100% packet loss, time 5106ms
```



允许特定主机SSH连接

启动服务器的SSH服务

启动服务，使得客户端可以通过22端口远程连接

```
(kali㉿kali)-[~]
$ /etc/init.d/ssh start 打开ssh服务
Starting ssh (via systemctl): ssh.service.

(kali㉿kali)-[~]
$ /etc/init.d/ssh status 查看ssh服务器状态
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/lib/systemd/system/ssh.service; disabled; vendor preset: disabled)
   Active: active (running) since Mon 2022-10-24 03:50:44 EDT; 37s ago
     Docs: man:sshd(8)
           man:sshd_config(5)
   Process: 3906 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
    Main PID: 3907 (sshd)
      Tasks: 1 (limit: 2251)
     Memory: 1.9M
        CPU: 53ms
    CGroup: /system.slice/ssh.service
            └─3907 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

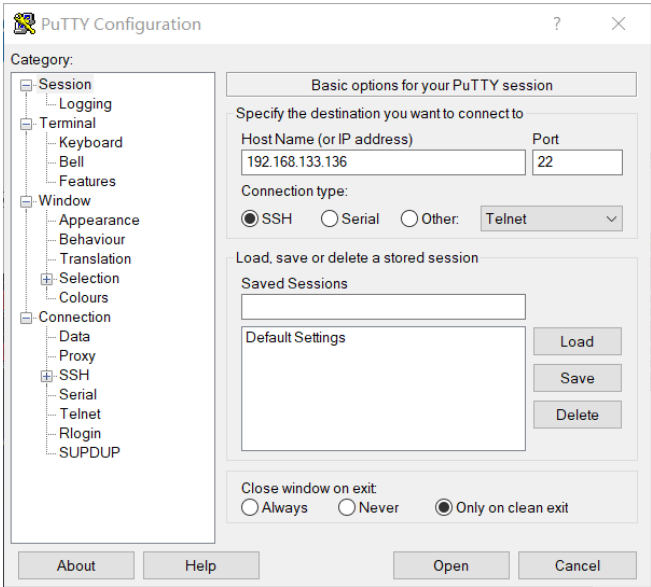
Oct 24 03:50:44 kali systemd[1]: Starting OpenBSD Secure Shell server ...
Oct 24 03:50:44 kali sshd[3907]: Server listening on 0.0.0.0 port 22.
Oct 24 03:50:44 kali sshd[3907]: Server listening on :: port 22.
Oct 24 03:50:44 kali systemd[1]: Started OpenBSD Secure Shell server.
```

客户端1、2分别SSH连接服务器

```
ubuntu@ubuntu-virtual-machine:~$ ssh kali@192.168.133.136
kali@192.168.133.136's password:
Linux kali 5.18.0-kali5-amd64 #1 SMP PREEMPT_DYNAMIC Debian 5.18.5-1kali6 (2022-07-07) x86_64

The programs included with the Kali GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Kali GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Mon Oct 24 04:03:29 2022 from 192.168.133.1
(kali㉿kali)-[~]
```





允许特定主机SSH连接

设置服务器的iptables规则

设置iptables规则，只允许特定远端主机SSH连接。

命令示例：iptables -I INPUT -p tcp --dport 22 -s 192.168.133.134 -j ACCEPT

iptables -A INPUT -p tcp --dport 22 -j DROP

```
(root@kali)-[~]
# iptables -L
Chain INPUT (policy ACCEPT)
target      prot opt source                destination
ACCEPT      tcp  --  192.168.133.134        anywhere        tcp dpt:ssh
DROP        tcp  --  anywhere               anywhere        tcp dpt:ssh

Chain FORWARD (policy ACCEPT)
target      prot opt source                destination

Chain OUTPUT (policy ACCEPT)
target      prot opt source                destination
```




允许特定主机SSH连接

验证防火墙效果

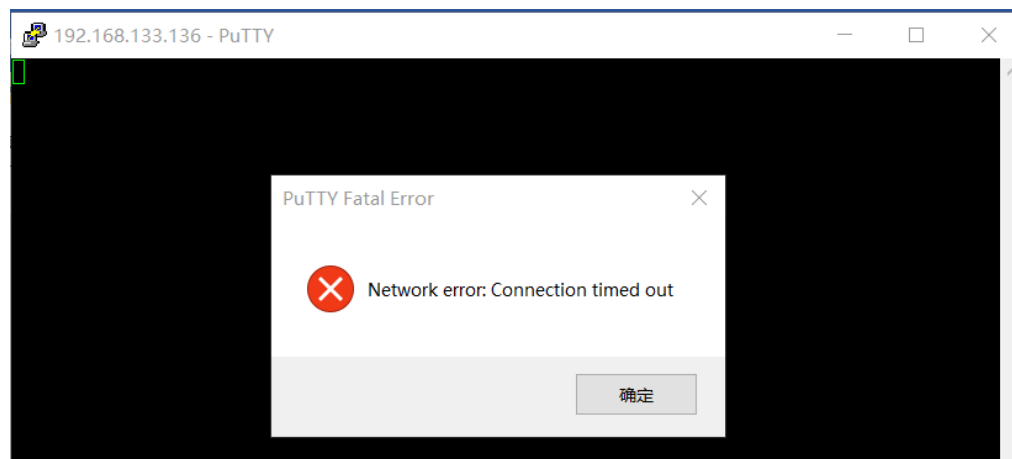
在iptables中被允许连接的客户端1仍可远程SSH连接

客户端2已无法连接到服务器

```
ubuntu@ubuntu-virtual-machine:~$ ssh kali@192.168.133.136
kali@192.168.133.136's password:
Linux kali 5.18.0-kali5-amd64 #1 SMP PREEMPT_DYNAMIC Debian 5.18.5-1kali6 (2022-07-07) x86_64

The programs included with the Kali GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Kali GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Mon Oct 24 04:03:29 2022 from 192.168.133.1
(kali)~$
```



02

NAT部分

- 设置iptables规则，实现内网http服务的发布



发布内网服务



防火墙

Ubuntu



内网主机PC1

Windows XP



公网主机PC2

Windows 10

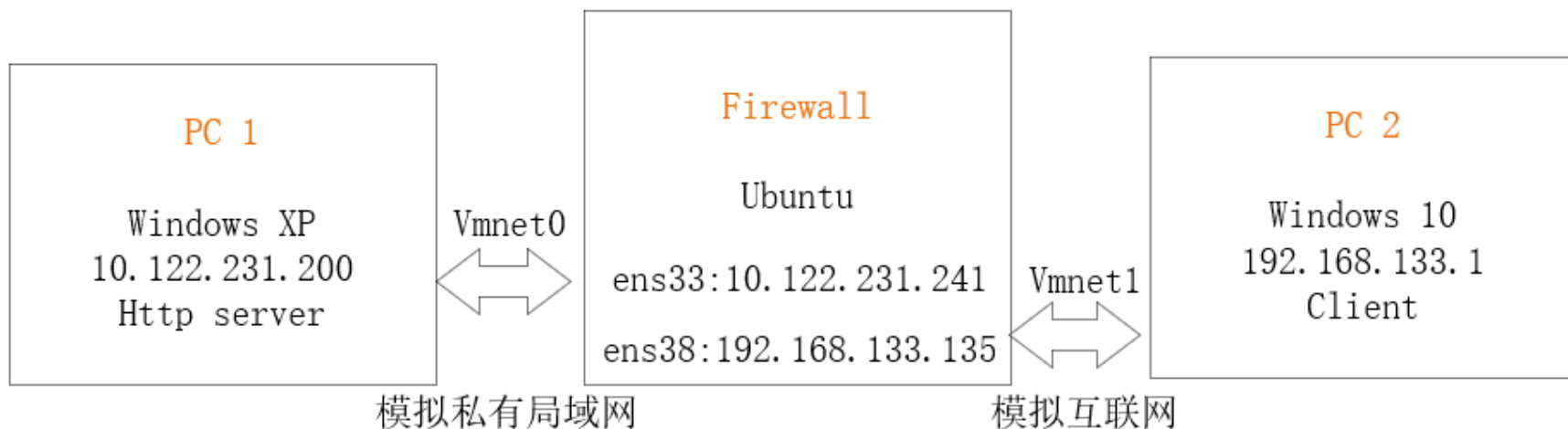


软件

iptables、nginx

了解实验场景

PC2不能直接访问局域网内的电脑PC1上的服务，但PC2可以与firewall上的公网IP相互通讯，由此可以进行NAT的设置，让PC2访问firewall的公网地址上的服务，再由firewall对数据进行处理，将访问请求发送到内网PC1上。





发布内网服务

配置网络环境

给作为防火墙的主机增设网卡，两张网卡使用不同的网络连接方式。

PC1和PC2分别与防火墙的两张网卡的连接方式相对应，同时将PC1的网关设置为部署了防火墙的主机。

虚拟机设置

硬件 选项

设备	摘要
内存	4 GB
处理器	2
硬盘 (SCSI)	20 GB
CD/DVD (SATA)	自动检测
网络适配器	自定义 (VMnet0)
网络适配器 2	自定义 (VMnet1)
USB 控制器	存在
声卡	自动检测
打印机	存在
显示器	自动检测

设备状态

- ☒ 已连接(C)
- ☒ 启动时连接(O)

网络连接

- ☐ 桥接模式(B): 直接连接物理网络
 - ☐ 复制物理网络连接状态(P)
- ☐ NAT 模式(N): 用于共享主机的 IP 地址
- ☐ 仅主机模式(H): 与主机共享的专用网络
- ☒ 自定义(U): 特定虚拟网络
 - VMnet1 (仅主机模式)
- ☐ LAN 区段(L):

LAN 区段(S)... 高级(V)...

```
ens33: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.122.231.241 netmask 255.255.192.0 broadcast 10.122.255.255
    inet6 fe80::1aed:1c18:4ed:4207 prefixlen 64 scopeid 0x20<link>
    ether 00:0c:29:f0:06:e4 txqueuelen 1000 (Ethernet)
    RX packets 1008 bytes 96986 (96.9 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 8455 bytes 660574 (660.5 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

ens38: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.133.135 netmask 255.255.255.0 broadcast 192.168.133.255
    inet6 fe80::8a20:1000:ae29:99d7 prefixlen 64 scopeid 0x20<link>
    ether 00:0c:29:f0:06:ee txqueuelen 1000 (Ethernet)
    RX packets 12 bytes 1382 (1.3 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 86 bytes 9101 (9.1 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

网络连接

本地连接 状态

常规 支持

连接

状态: 已连接上

持续时间: 01:37:54

速度: 1.0 Gbps

活动

发送 收到

字节: 2,145,094 1,980,832

属性(E)

禁用(D)

本地连接 属性

常规 高级

连接时使用:

VMware Accelerated AMD PCNet /

配置...

此连接使用下列项目:

- ☒ Microsoft 网络客户端
- ☒ Microsoft 网络的文件和打印机共享
- ☒ QoS 数据包计划程序
- ☒ Internet 协议 (TCP/IP)

安装(A)... 卸载(U) 属性(E)

说明

TCP/IP 是默认的广域网协议。它提供跨越多种互联网网络的通讯。

☐ 连接后在通知区域显示图标(I)

☒ 此连接被限制或无连接时通知我(M)

Internet 协议 (TCP/IP) 属性

常规

如果网络支持此功能，则可以获取自动指定的 IP 设置。否则，您需要从网络系统管理员处获得适当的 IP 设置。

☐ 自动获得 IP 地址(A)

☒ 使用下面的 IP 地址(S):

IP 地址(I): 10.122.231.200

子网掩码(M): 255.255.192.0

默认网关(G): 10.122.231.241

☐ 自动获得 DNS 服务器地址(A)

☒ 使用下面的 DNS 服务器地址(S):

首选 DNS 服务器(P):

备用 DNS 服务器(A):

Ubuntu作为网关



发布内网服务

测试网络连通性

完成网络环境设置后，应实现：

- PC1可以和防火墙互相ping通；
- PC2可以和防火墙互相ping通；
- PC1和PC2互ping不通

开启防火墙的路由转发，设置NAT规则

命令示例：

```
echo 1 > /proc/sys/net/ipv4/ip_forward
```

```
iptables -t nat -A PREROUTING -i ens38 -p tcp -d 192.168.133.135 --dport 80 -j DNAT --to-destination 10.122.231.200
```

```
iptables -t nat -A POSTROUTING -o ens38 -p tcp -j SNAT --to-source 192.168.133.135
```

```
iptables -A FORWARD -i ens38 -p tcp -d 10.122.231.200 --dport 80 -j ACCEPT
```

```
iptables -A FORWARD -p tcp -m state --state established,related -j ACCEPT
```



发布内网服务

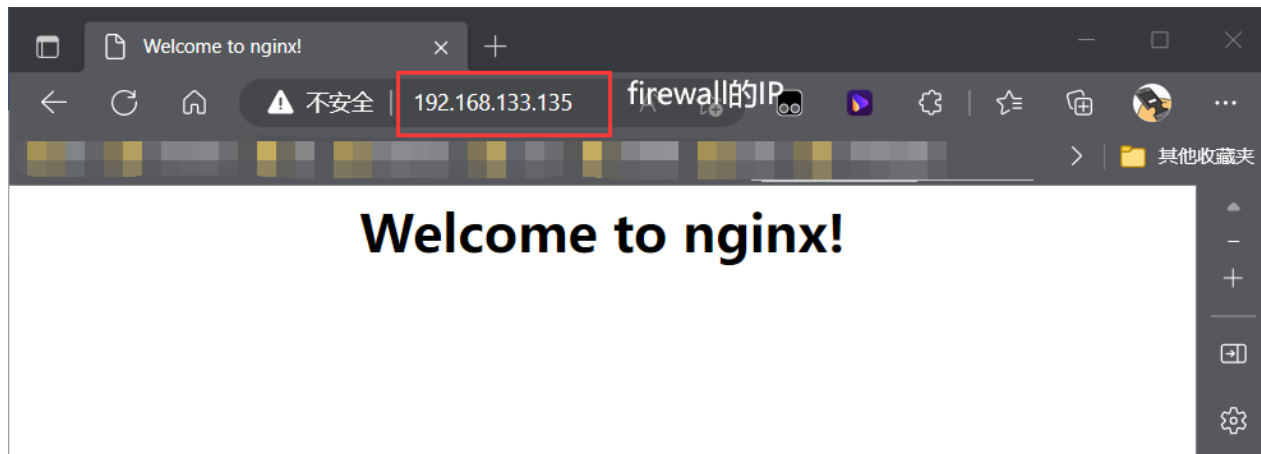
公网主机PC2使用防火墙IP访问内网服务

启动PC1的http服务后，PC2可以通过使用防火墙公网IP访问内网服务

```
C:\nginx-1.1.0>start nginx
C:\nginx-1.1.0>netstat -an

Active Connections

Proto Local Address           Foreign Address         State
TCP    0.0.0.0:80               0.0.0.0:0               LISTENING
TCP    0.0.0.0:135              0.0.0.0:0               LISTENING
TCP    0.0.0.0:445              0.0.0.0:0               LISTENING
TCP    10.122.231.200:139       0.0.0.0:0               LISTENING
TCP    127.0.0.1:1026           0.0.0.0:0               LISTENING
TCP    127.0.0.1:8182           0.0.0.0:0               LISTENING
```





谢谢！